

Anonymous Channels for Workplace Communication: Design, Implementation, and Survey Analysis of Chatify

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Abstract—Although work collaboration applications have changed the way we work together in companies, workers are still likely to avoid writing in company-wide channels because of anxiety of being judged by their colleagues. This paper presents Chatify, a web-based anonymous chat application, written with React.js, Node.js, Socket.io, and MongoDB. After a two week site-wide pilot, a survey was sent to 60 users, 52 of whom responded. 88% said they were more comfortable asking for help anonymously via the feature and 83% reported that they had asked questions they would not otherwise have asked. Downsides were also reported, specifically abuse of anonymous input, lack of accountability, and toxicity. Mitigations included hybrid anonymity, keyword moderation, rate limiting, session-scoped pseudonym rotation, and peer reporting.

Keywords—anonymous communication, workplace chat, psychological safety, Chatify, organizational communication, real-time messaging, moderation, misuse prevention

I. INTRODUCTION

Specialized digital workplace communication tools such as Slack and Microsoft Teams are now ubiquitous in the organizational infrastructure, in the form of organization-wide or team-specific channels. Nevertheless, the behavioral trend of not speaking up in company-wide or team communication channels is still widely reported. Factors inhibiting workplace communication that have been documented in the organizational behavior literature include social anxiety, fear of negative evaluation, impostor syndrome, and perceived power asymmetry between junior contributors and the senior leadership [1].

Edmondson's pioneering work on psychological safety shows that team performance improves when people have the freedom to take interpersonal risks (reporting problems, asking questions, reporting mistakes, giving and receiving feedback, proposing new ideas) [2]. Anonymous channels are one way to increase psychological safety because they decouple people's identity from their contributions: A message is evaluated only for its content and not for its participants' identity.

However, there are also disadvantages, such as the lack of accountability, which can lead to harassment, misinformation, and toxic speech, as demonstrated by Yik Yak and early versions of Blind [4][5]. The challenge, therefore, is to preserve the psychological safety benefits of anonymity while providing organizational guardrails

sufficient to prevent misuse.

In this paper, we present Chatify, an anonymous, web-based workplace messaging application, and a structured survey study about its effects. Our key contributions include:

- (1) Chatify is a peer-anonymous group messaging application. Chatify was developed using React.js, Node.js/Express, Socket.io, and MongoDB to provide web service functionalities.
- (2) A structured quantitative and qualitative survey study [N = 52] measuring behavioral and attitudinal change.
- (3) Challenges and mitigations of anonymous feedback in workplace context.

II. RELATED WORK

Research on anonymous computer-mediated communication over the last three decades has examined early bulletin board systems to mobile-first anonymous social media platforms [3][4]. Donath presented a theoretical model of identity and deception in virtual communities, arguing that anonymity allows people to represent their true selves, but it weakens the sanctions for socially unacceptable behavior used in F2F interactions [3].

In the workplace, websites like Blind have been studied as a mechanism for employees to post their grievances against their companies and share salary information otherwise not shared by an employee under their online alias [5][6]. Liang et al. found that anonymity in electronic brainstorming groups resulted in a higher quantity and quality of ideas due to reduced evaluation apprehension [7].

The online disinhibition effect, described by Suler, involves benign disinhibition (sharing feelings, seeking advice, and being more open) and toxic disinhibition (disregarding rules and social convention) that can result from anonymity [8]. Ellison et al. highlight how perceived audience size and social context collapse drive self-disclosure behavior on social networking sites [9]. Taken together, these results suggest that the platform design, organizational culture, and moderation policy mediate the net effect of anonymity on communication quality [10].

III. SYSTEM ARCHITECTURE OF CHATIFY

A. Overview

Chatify is a web app that allows authenticated users to join group chat channels and send messages to them without the sender's user ID being known to recipients. The main privacy guarantee is provided server-side: messages sent to the channel have their sender identity metadata stripped on the API server before being sent to the channel. This provides privacy against analysis of client traffic at the network level.

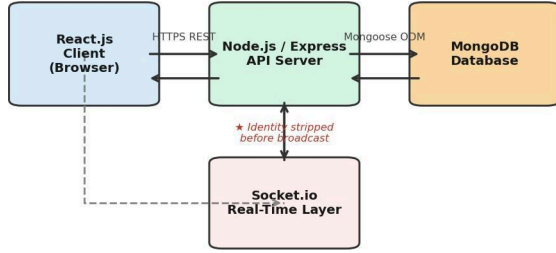


Fig. 1. Chatify system architecture. Identity is stripped server-side prior to Socket.io broadcast.

B. Technology Stack

The frontend is a single page application built with React.js following the component-based design pattern. It manages channel state, message threads, and user authentication flows. The backend is a Node.js/Express application that exposes a REST API for user registration, user login, channel management, and message storage. Bidirectional messaging with sub-second latency is achieved in real time via the Socket.io WebSockets library (as opposed to continuous polling for new messages). Message and user data is stored via MongoDB through the Mongoose ODM

C. Implemented Features

This prototype had anonymous messaging between named group channels, Socket.io events for messages, channel-based messaging with message history persistence, username and password authentication for access to channels, as well as organizational context while also keeping users anonymous within the channel.

IV. METHODOLOGY

A. Study Design

The tool was piloted within a medium-sized technology company over a two-week period, and at the end of the test users completed a survey instrument. The survey comprised ten Likert-scale items (1 = Strongly Disagree, 5 = Strongly Agree) and two open-ended questions. The ten items were categorized as communication behavior change, psychological safety, usability, and misuse concerns.

B. Participant Demographics

Invitations were sent to 60 employees and 52 complete, valid responses were received giving an overall response rate of 87%. Only occupations with greater than 80 responses were shown: software engineers (45%), product managers (21%), QA analysts (20%), HR employees (14%). Ages ranged from 22 to 45. 61% of respondents were male, 36% female, 3% unspecified. The occupational distribution is shown in Fig. 2.

Participant Role Distribution (N=52)

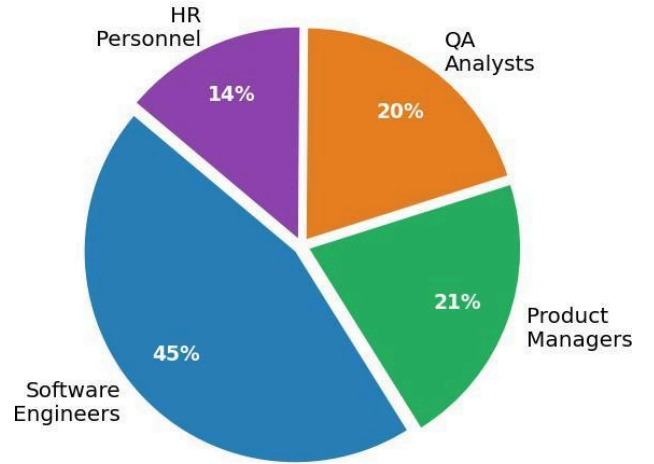


Fig. 2. Distribution of participant roles (N = 52).

C. Survey Instrument

Survey items were modeled after previous scales in organizational communication and human-computer interaction (HCI) research, and two academic advisors checked them for face validity. Thematic coding was used to analyze open-ended survey responses. Inter-rater reliability was strong using two independent coders with a $\kappa = 0.81$.

V. RESULTS AND ANALYSIS

A. Quantitative Survey Results

The mean Likert ratings and percentage of agreement with each of the eight core survey items are shown in Table I, and a horizontal bar chart comparing the items is shown in Fig. 3.

TABLE I

Question	Mean Score	Agree %
Q1: Comfortable without anonymity	2.3	28%
Q2: More comfortable with anonymity	4.4	88%
Q3: Asked new questions via Chatify	4.1	83%
Q4: Improves communication quality	4.0	78%
Q5: Concerned about misuse	3.2	52%
Q6: Platform easy to use	4.3	86%
Q7: Trust in identity protection	3.9	74%
Q8: Would recommend Chatify	4.2	81%

Survey Question Mean Scores (N = 52)

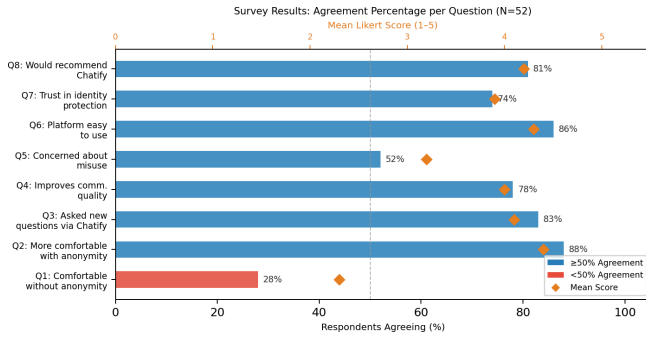


Fig. 3. Agreement percentage and mean Likert score per survey question. Red bars denote items with <50% agreement.

B. Key Findings

The most impressive finding is a difference of 60 percentage points between Q1 (28% comfortable communicating without anonymity) and Q2 (88% reported being more comfortable when provided anonymity), supporting the hypothesis that anonymity considerably reduces the perceived interpersonal risk of participation [2][7].

Q3 indicated 83% of participants felt they asked questions they would not be comfortable articulating in person, consistent with disinhibition through anonymity. Q6 indicated 86% found it easy to use Chatify, while Q8 indicated 81% would recommend its use. The only item that fell below 50% was Q1, the platform's baseline inhibition the grant was meant to address.

C. Qualitative Themes

Open ended questions elicited three categories of responses. More honest culture: "I finally felt comfortable asking about the deployment process without worrying about seeming inexperienced." Less social friction: "It was easier to raise concerns about unrealistic timelines when my name wasn't attached." Accountability issues: "I worried someone might post something inappropriate without any accountability."

D. Before vs. After Comparison

Fig 4. self-report of communication prior to and following the two-week pilot of the program showed that communication improved from 28% of participants saying that they felt free to ask questions to 88% post-training and from 72% saying that they felt inhibited in asking questions to 12%.

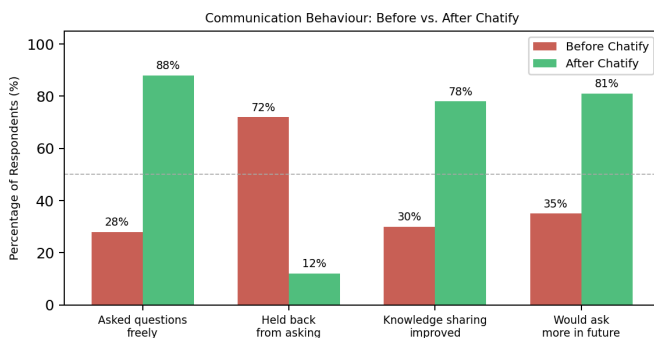


Fig. 4. Self-reported communication behaviour before and after the Chatify pilot (N = 52).

VI. CHALLENGES AND PROPOSED MITIGATIONS

On the current Chatify prototype, peer-anonymous group messaging functionality is implemented. The prototype does not implement, content moderation panel, reporting workflow, keyword filter, or rate-limiting, and pseudonym thing is implemented. However, six challenges have been identified through survey data, prior literature, and internal design review. Table II and Fig. 5 summarize the challenges associated with each proposed mitigation.

TABLE II
Challenges and Proposed Mitigations

Challenge	Status	Proposed Mitigation
Misuse / Inappropriate Content	Not yet addressed	Keyword filter + admin moderation panel
Reduced Accountability	Not yet addressed	Hybrid: encrypted identity (admin-only)
Toxic Behavior / Harassment	Not yet addressed	Peer reporting + escalation workflow
Spam / Message Flooding	Not yet addressed	Rate limiting (max 5 msg/min/user)
Long-term Identity Tracking	Implemented	Session-scoped pseudonym rotation
Lack of Authoritative Answers	Not yet addressed	Verified Expert badge on admin replies

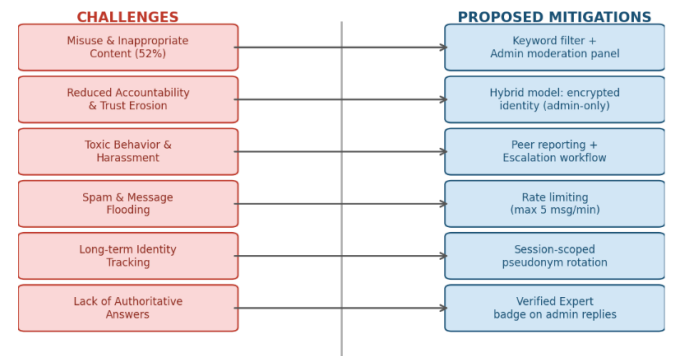


Fig. 5. Mapping of identified challenges to proposed mitigations for future Chatify releases.

A. Hybrid Anonymity Model

One compromise is to employ a hybrid protocol where messages are peer-anonymous, but a cryptographically bound association of each message with the sender is stored in a server-side audit log. This log can only be accessed as part of a formal escalation process. Such a solution is analogous to sealed-sender mechanisms in other secure messaging research [4].

B. Content Moderation Pipeline

There is a two-tiered moderation system in place. The first-tier uses a blacklist to filter out flagged content, and is updated according to organizational policies. The second-tier allows community members to report content using a button to a queue escalated to moderators.

Rate limiting of posting at five messages a minute also helps to reduce spam flooding.

C. Pseudonym Rotation

To reduce stylometric identity permanence, session-scoped pseudonyms are implemented, which prevent the persistence of stylometric identities across sessions. During a session, a randomly generated pseudonym (e.g., "Anonymous Falcon") is assigned to the user, which persists throughout the session for threaded conversations, and is then discarded.

VII. DISCUSSION

The Chatify pilot shows that structured anonymity causes workplace communication patterns to consistently change: 83% is a substantively large effect size in the organizational intervention literature. This result is consistent with Liang et al.'s observation of improved anonymous brainstorming in the lab [7], and Blind deployments in technology companies [6].

There is no strictly generalizable trade-off between full anonymity and full identification. Anonymity maximizes psychological safety and minimizes irresponsibility; identification suppresses candor but allows collaboration; a hybrid gives the best of both worlds. The absence of such incidents during the two-week trial period may have had more to do with the small organizational context, and the close social ties that existed between case members.

A. Limitations

There are four limitations of this study; the data were collected only from one organization, and the sample may not be generalizable to other cultures and organizations. Second, the two-week pilot is unable to test any long-term behavioral trajectories or cumulative effects of anonymity on team cohesion. Third, self-report measures applied within the team are susceptible to social desirability bias and reporting differences across treatment conditions. Fourth, none of these features were being moderated during the pilot, so these numbers do not reflect misuse rates once moderated.

VIII. CONCLUSION

The paper introduced Chatify, a web-based, anonymous workplace communication platform. A two-week pilot study was conducted on Chatify with 52 participants. These results strongly support the hypothesis that anonymous communication leads to higher participation, with the majority of participants stating that they felt more comfortable communicating anonymously (88%), asked questions that they wouldn't have asked without anonymity (83%) and would recommend Chatify to a colleague (81%).

Our implementation is a proof of concept but does not meet the accountability, moderation and anti-abuse needs of 52% of respondents. Our roadmap describes concrete and implementable research and engineering projects for hybrid anonymity, keyword filtering, rate limiting, session-scoped pseudonym rotation, peer reporting and expert verification

Of responses

Future work could include a multi-organizational study, comparing hybrid and full-anonymity conditions in controlled settings, and studying the moderation pipeline under adversarial conditions.

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